module-07-eleven-multiplication - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 7 — ×11 Multiplication (Darśana)

Sanskrit: *Ekādhikena Pūrveṇa* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Mental multiplication by 11 from 20th-century Indian arithmetic.

> **Darśana note.** This workshop condenses Module 7's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Project: 234 × 11 = ? Pause. *"Long multiplication takes ~30 seconds. The Vedic Mathematics method takes ~5 seconds. Today we learn the method AND examine its honest historical claim — manuscript date 1911-1918, published 1965."*

Core idea · 15 min

×11 mental method (Vedic Mathematics by Bharati Krishna Tirthaji, 1965).

Algebraic identity: (10a + b)(10 + 1) = 100a + 10(a+b) + b.

Procedure: 1. Split 2-digit number into a and b. 2. Cross-add: a + b. 3. Place: a, (a+b), b. Carry if needed.

Extension to 3-digit: abc × 11 = a, (a+b), (b+c), c.

Honest historical caveats: - Tirthaji's manuscript: 1911-1918. - Published posthumously: 1965. - Claimed source: a Parisistha of the Atharvaveda. **This manuscript has never been independently verified.** - Mathematician S.G. Dani (1993) critique: the methods are 20th-century arithmetic, not Vedic.

Use the method; teach the honest history.

Demonstration · 10 min

Algebra explainer + carry-case worked example. Derive $(10a + b) \times 11 = 100a + 10(a+b) + b$ on the board with the carry rule. Then demo $99 \times 11 = 1089$ (the most carry-heavy case). Discuss generalization to 3-digit.

Source moment · 5 min

Quoted reference. Tirthaji, *Vedic Mathematics* (1965), Sūtra 1 — *Ekādhikena Pūrveṇa*:

> "By one more than the previous one."

This is one of 16 sūtras. The phrase itself does not appear in extant Atharvaveda manuscripts. Treat as 20th-century synthesis.

Hands-on activity · 15 min

Historical-method critique. Each student writes 250 words on: *"Tirthaji's ×11 method works. Its claimed Vedic origin is unverified. How would you teach this honestly to (a) a curious child, (b) a sceptical adult?"*

Reflection + take-home · 10 min

"This module taught you a technique and its honest historical context. How does separating method from origin claims serve mathematical education?"

Go deeper into Adhyayana

The 10 days you're skipping if you only do this workshop:

- **Day 01** — The ×11 hook — what is the trick?
- **Day 02** — The carry case
- **Day 03** — Why it works — the algebra
- **Day 04** — 3-digit ×11
- **Day 05** — Speed day
- **Day 06** — Anurūpyena extension to ×12-19
- **Day 07** — Limits of the method
- **Day 08** — Tirthaji 1965 and the historical claim
- **Day 09** — Mixed practice
- **Day 10** — Final assessment

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.